

POWER MANAGEMENT

TODO: CHECK POLARITY OF 3-PIN JST

J1 Conn_01x03 JST 3pin Battery terminal

BATT_NTC

VBAT

GND

VBAT

R33 1M

C16 100nF

VBAT_SENSE

Voltage divider 2:1 For RAK4630 AIN2/P0.04 Backup Voltage measurement

R34 1M

GND

GND

VBUS_5V

C11 1uF

PWR_FLAG

R6 10k

GND

GND

VCC_3v3

R2 1k

GND

R3 1k

GND

GND

U1 BQ24074RGT

ITEM IN

OUT

CE

TMR

EN1

EN2

ILIM

ISRT

VSS

BAT

TS

PGOOD

CHG

VBAT

BATT_NTC

R1 10k

VCC_3v3

U3 MAX17048G_T10

CTG

CELL

VDD

EP

QSTRT

SCL

SDA

'Fuel gauge'

Battery level monitor

VBAT

PWR_FLAG

BATT_ALERT

GND

GND

GND

BATT_I2C_SDA

BATT_I2C_SCL

R5 4.7k

R4 4.7k

VCC_3v3

VCC_3v3

C9C 1

VBAT Regulator (LiPo 3.0V-4.2V to stable VCC_3.3v for system)

U2 TPS62840DLCR

VIN

EN

MODE

STOP

VSET

SW

VDS

PWR_FLAG

L1 2.2uH

VCC_3v3

C1 C 10uF

VBAT

GND

VBAT

GND

GND

GND

C3 C 10uF

C2 C 10uF

GND

The schematic illustrates the RAK4630 (NRF52840 MCU + SX1262 LoRa RFIC) board layout. Key components and connections include:

- Central Chip:** RAK4630 (NRF52840 MCU + SX1262 LoRa RFIC) with pins 1 through 44 labeled.
- Antenna:** AE1 NFC Antenna 13.56mhz connected to pins 13 and 14.
- Power Regulation:** VBAT and VBUS_5V inputs, with decoupling capacitors (C4, C5, C6, C7, C10) and resistors (R7, R21, R22, R23, R25, R26).
- USB-C Connector:** J3 USB_C_Receptacle_USB2.0_14P connected to the USB-C shield.
- Control and Status:** BZ1 Buzzer, D2 LED, SW1 SW_Push, SW2 SW_Push, and status LEDs (JP2, JP3) connected to various pins.
- Flash Memory:** U5 W25Q128 qSPI Flash memory placeholder, crossed out with a red X, indicating it is not to be populated.
- Test Points:** TP14, TP15, TP16, TP17, TP18, TP20, TP21, TP22, TP23, TP24, TP25, TP26, TP27, TP28, TP29, TP30, TP31, TP32, TP33, TP34, TP35, TP36, TP37, TP38, TP39, TP40, TP41, TP42, TP43, TP44, TP45, TP46, TP47, TP48, TP49, TP50, TP51, TP52, TP53, TP54, TP55, TP56, TP57, TP58, TP59, TP60, TP61, TP62, TP63, TP64, TP65, TP66, TP67, TP68, TP69, TP70, TP71, TP72, TP73, TP74, TP75, TP76, TP77, TP78, TP79, TP80, TP81, TP82, TP83, TP84, TP85, TP86, TP87, TP88, TP89, TP90, TP91, TP92, TP93, TP94, TP95, TP96, TP97, TP98, TP99, TP100.

Monarch 2 GM02SP Module (CAT-M/NB-IoT + GNSS Receiver)

0mm pitch wire connection

Conn_01x08

J5

TP19 GND

VBAT

GND

UART0_RX

UART0_TX

PS_STATUS

WAKED

RESETN

LTE-ANT

R18 OR

R17 OR

R16 OR

J7 CONUFL001-SMD-T

SIG

GND_2

GND_1

GND

GNSS-ANT

R27 OR

R24 OR

R28 OR

J8 CONUFL001-SMD-T

SIG

GND_2

GND_1

GND

IC2 GM02SP

TP13 GM02SP_VBAT

VBAT

GND_12

VBAT_2

VBAT_3

GND_13

GND_14

JTAG_TMS

JTAG_TDI

JTAG_TDO

JTAG_TCK

JTAG_TRSTN

DNC_4

ADC1

SIMO_VCC

DNC_5

DNC_6

DNC_7

DNC_8

DNC_9

GND_15

GND_16

GND_17

GND_18

GND_19

GND_20

GND_21

GND_22

GND_23

GND_24

GND_25

GND_26

LTE_ANT

GND_27

GND_28

GND_29

DNC_9

GPI023/_I2C_SDA

GPI04/_PCM_CLK

GPI024/_I2C_SCL

GPI03/_PCM_RXD

GPI05/_PCM_FS

GPI06/_PCM_TXD

GND_30

GND_31

GND_32

GND_33

GND_34

GND_35

GND_36

GND_37

GND_38

GND_39

GND_40

GND_41

GND_42

GND_43

GND_44

GND_45

GND_46

GND_47

GND_48

GND_49

GND

TP7 JTAG_TMS

TP9 JTAG_TDI

TP10 JTAG_TDO

TP11 JTAG_TCK

TP12 JTAG_TRSTN

VSIM

TP5 RES_FFF/FFH

R11 100k

GND

TP6

TP8 RXD2

TP1 TXD2

TP2 RTS1

TP3 RXD1

TP4 CTS1

TXD1

UART0_RX

UART0_TX

TP26 UART0_TX

TP27 UART0_RX

TP28 GND

SIM_CLK

SIM_RSET

SIM_IO

SIM_DETECT

RESETN

WAKED

VCC_3v3

GM02SP

GND_1

GND_2

GND_3

GND_4

GND_5

GND_6

GND_7

GND_8

GND_9

GND_10

GND_11

VBAT_1

GPI09/_SPI_CLK

GPI07/_SPI_SDI

GPI08/_SPI_SDO

GPI010/_SPICSN1

GND_2

GPI011/_SPICSN2

1V8

GPI033/_TX_IND

GPI034/_ANT_TUNE0

GPI035/_ANT_TUNE1

GPI01/_STATUS_LED

GPI02/_PS_STATUS

GPI028/_DTR0

DNC_1

RING0

DNC_2

GPI032/_PWM1/_PULSE1

GPI031/_PWM0/_PULSE0/_19M2_OUT

RESERVED/_FFF_FFH

GPI021/_CTS3

GPI020/_RXD3

GPI022/_RTS3

GPI019/_TXD3

GPI018/_RTS2/_DSR0

GPI016/_RXD2

GPI017/_CTS2/_DCD0

GPI015/_TXD2

RTS1

RXD1

CTS1

TXD1

RTS0

GPI013/_RXD0

GPI014/_CTS0

GPI012/_TXD0

GND_3

GND_4

GPI025/_SIM1_IO

GPI026/_SIM1_CLK

GPI027/_SIM1_RESETN

SIMO_CLK

SIMO_RSTN

SIMO_IO

SIMO_DETECT

RESETN

WAKE1

WAKE0

WAKE2

GND_5

GND_6

GND_7

DNC_3

GND_8

WAKE3

GND_9

WAKE4

GND_10

GND_11

VBAT_1

R134 7k

VCC

RST

CLK

GND

VPP

I/O

CD

SHIELD

SHIELD_1

J6 SIM8055-6-1-14-00-A (NanoSim card holder socket)

VSIM

1V8_OUT

NX3008PBKM8 T1

R14 100k

SIM_DETECT

R15 100k

GND

GND

GND

C12 100nF

TOOD:

--check jst polarity

--check GNSS pin 94, if not, may need to change GM02SP symbol/footprint -- bailache.

-- Investigate if LED2 (p134) can be used for PWM for passive buzzer melody generation

--also same for IC 2 pins; can they be repurposed? how 7 pin definitions in code?

Sheet: /

File: BPV4_Collar.kicad_sch

Title:

Size: A3

Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 1/1